

Understanding and Modeling Organizational Systems

CSE 4407

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What is an Organization... as a System?

- **Core Idea:** Organizations are systems designed to achieve goals using resources (people, tech, etc.)
- **Structure:** Composed of smaller, interrelated **subsystems** (departments, units)
 - **Example:** Accounting, Marketing, Production, IT, Operations, Legal
- **Integration:** Subsystems work together to form the whole organization
- **Key Benefit:** Systems principles help us understand how organizations *really* work

The Web of Connections: Interrelatedness

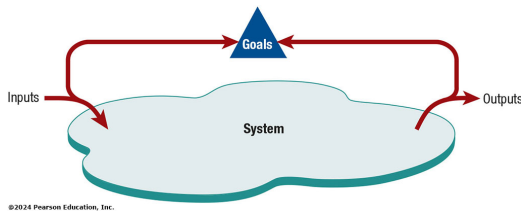
- **Fundamental Truth:** All systems and subsystems are **interrelated** and **interdependent**
- **Ripple Effect:** Changing one element affects the *entire* system
 - Removing admin assistants and replacing them with PCs impacts
 - The assistants (obviously!)
 - Managers (new tasks?)
 - Anyone who relied on the assistants' informal communication networks
- **The takeaway for analysts:** Always consider the wider impact of system changes

System Mechanics: Inputs, Processes, Outputs, and Boundaries

- **Input:** Resources from the environment (people, materials, information)
- **Process:** Transformation of inputs (verifying, updating, calculating, manufacturing)
 - *Is something being changed? If not, it might not be a process*
- **Output:** Results returned to the environments (products, services, information)
- **Boundaries:** Separate the system from its environment
 - Can be **permeable** (open, easy exchange) or **impermeable** (closed, restricted exchange)
 - Organizations need *some* permeability to survive (import resources, export products)

Staying on Course: Feedback and Environment

- **System Control:** Organizations use planning and control
- **Feedback:** Output is compared to goals to guide future inputs/actions
 - **Example:** Poor sales of red, white, and blue weight sets (output) leads to producing fewer (planning/control)
- **Ideal System:** Self-correcting/self-regulating
 - Inventory system automatically adjusts production/dyeing based on real-time sales data (Italian knitwear co.)
- **Environment:** Everything outside the boundary. Crucial influences:
 - Community (demographics, location)
 - Political (government regulations)
 - Economic (markets, competition)
 - Legal (laws, guidelines)

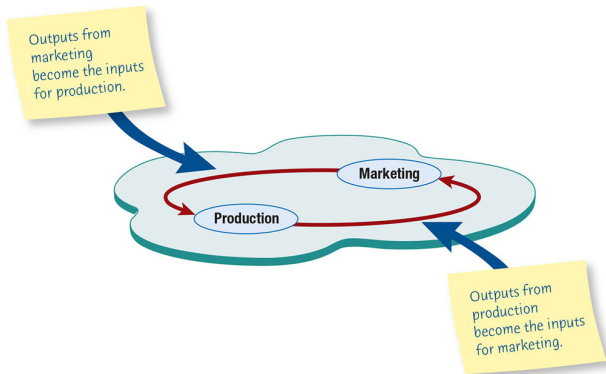


Open vs. Closed and The Rise of Virtual

- **Openness/Closedness:** Exists on a continuum internally too
 - Refers to flow of information/interaction *within* the organization
 - No system is completely open or closed
- **Virtual Organizations/Teams**
 - Not tied to a single physical location
 - Use technology (networks, communication tools) to connect people
 - Can adapt quickly to changing needs
 - **Benefits:** Reduced facility costs, rapid response, work-life balance [1]
 - **Challenges:** Maintaining culture, social connection (e.g, the need for university swag!)
 - Many System Analysis/Design teams work virtually

The Power of Perspective

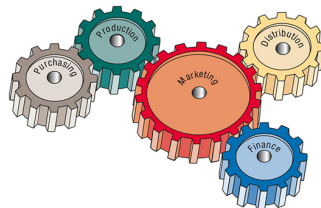
- **Systems Perspective:** Allows analysts to understand the *whole* business
- **Crucial Insight:** Subsystems *rely* on each other. Their work is interrelated
- **The Danger:** Managers often see the organization through the lens of their *own* department



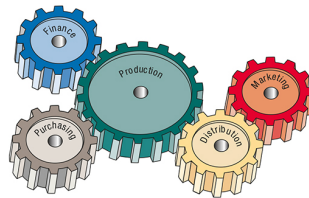
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Warped Views: Managerial Perspectives

- **The Problem:** Managers may overemphasize their own function's importance
- **Implication:** If these managers rise to strategic levels, their biased view can lead to poor decisions for the overall organization



How a Marketing Manager May View the Organization



How a Production Manager May See the Organization
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Integrating the Organization: ERP Systems

- **Concept:** Enterprise Resource Planning (ERP) systems aim to integrate information flow across function areas
 - **Example:** SAP, Oracle Cloud
- **Goal:** Provide a unified view of the organization's data and processes
- **Approach:** Usually purchased software, then customized
- **Key Difference:** Often requires changing business processes to fit the software (unlike traditional analysis which designs processes first)
- **Analyst Role:** Managing interfaces between ERP and existing ('legacy') systems

ERP: The Game Changer

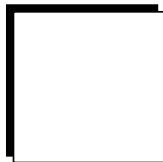
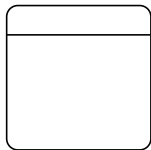
- **Complexity:** ERP implementations are massive projects impacting *everything*
- **Key Hurdles**
 - User acceptance (getting people to use it correctly)
 - Integrating with legacy systems and supply chain partners
 - Upgrading complex modules
 - Reorganizing work and decision-making
 - Expanded reach (connecting multiple organizations)
- **Impact:** Changes job design, required skills, and even company strategy
- **Long-term View:** Can make employees more effective, but requires careful planning, especially regarding user experience

Visualizing the System: Graphical Models

- **Why:** To understand and communicate:
 - System boundaries (what's *in*, what's *out*)
 - Information flow and usage
 - Relationships between system components
- **Think:** Blueprints and Maps for the System
- **Two Key Early Models**
 - Context-Level Data Flow Diagram (DFD)
 - Entity-Relationship (E-R) Model

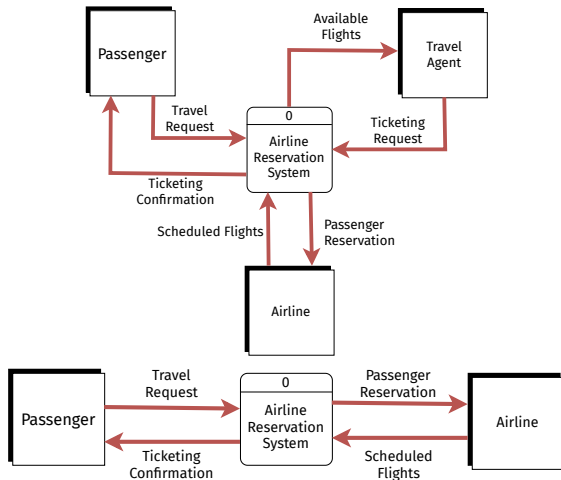
Zooming Out: The Context-Level DFD

- **Purpose:** Shows the system's *context* – how it interacts with the outside world
- **Think:** A higher-level “bird’s-eye view”
- **Focus:** Data flowing IN and OUT, and the overall system process
- **Key Components:** Only 3 Symbols!
 - **Process:** The entire system (rounded rectangle). *Transforms data.*
 - **External Entity:** Sends/Receives data (shaded square). *Outside the system.*
 - **Data Flow:** Movement of data (arrow)



Context-Level DFD Example: Airline Reservations

- Shows the system boundary and interactions clearly
- Example
 - **Before Online Booking:** Passenger and Travel Agent interact with the System; System interacts with Airline
 - **After Online Booking:** Travel Agent entity removed; Passenger interacts directly
- Illustrates how context changes affect the diagram



Context-Level DFD: Scope and What It Doesn't Show

- **Strengths**

- Excellent for defining **scope** (what's included vs. excluded)
- Shows key external interactions
- Easy to understand and create

- **Limitations**

- Shows *no* internal details (the “how”)
- Doesn't show data storage
- Data flows are high-level (e.g., “Passenger Reservation” - details hidden)

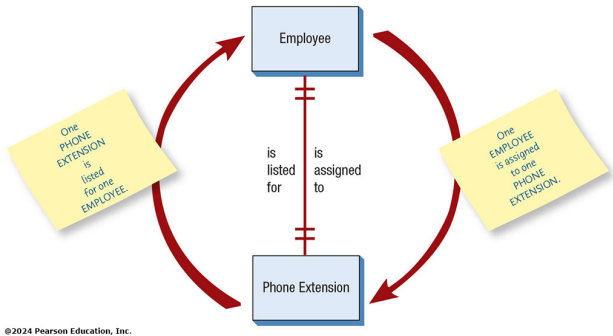
- **Use:** The very first step in understanding system boundaries. A prerequisite for more detailed diagrams

Modeling Data Structures: Entity-Relationship (E-R) Models

- **Alternative View:** Focuses on the *data* the system needs, not the *processes*
- **Core Components**
 - **Entities (Rectangles):** People, places, objects, events, or concepts about which data is stored (e.g., CUSTOMER, PRODUCT, ORDER)
 - **Relationships (Lines connecting entities):** How entities are associated or interact (e.g., a CUSTOMER *places* an ORDER)
- **Purpose**
 - Defines the structure of the data required by the organization/system
 - Foundation of database design

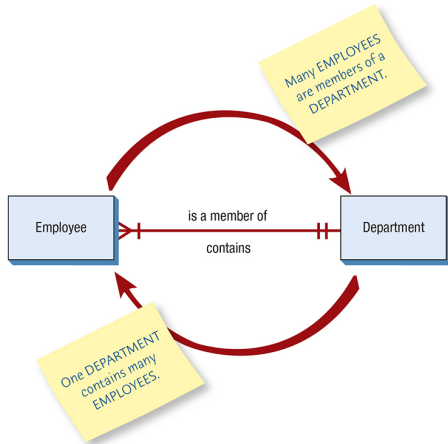
E-R Basics: Reading Relationships (1:1)

- **Cardinality:** Specifies *how many* instances of one entity relate to *how many* instances of another
- **Example:** One-to-One (1:1)
 - One EMPLOYEE is assigned to exactly one PHONE EXTENSION
 - One PHONE EXTENSION is listed for exactly one EMPLOYEE
 - || (two short parallel marks) often means 'exactly one'



E-R Relationships: One-to-Many (1:M)

- **Example:** One-to-Many (1:M)
 - Many EMPLOYEEs are *members of* one DEPARTMENT (Many side)
 - One DEPARTMENT *contains* many EMPLOYEEs (One side)
- **Notation**
 - > | (Crow's foot) often means 'many' (zero, one, or more)
 - || still means 'exactly one'
- Read the relationship name *towards* the entity you're going to

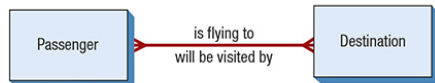
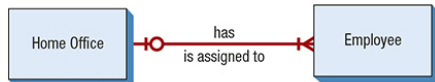
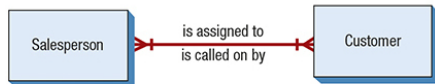
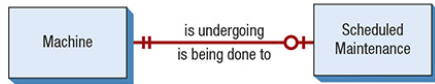
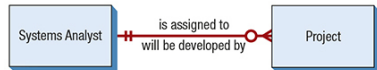
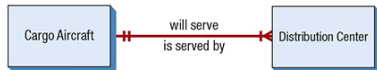
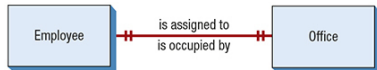


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E-R Notation: Cardinality and Optionality

- Common Crow's Foot Symbols

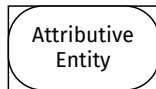
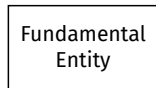
- ||: Exactly One
- > |: One or More (Many)
- o |: Zero or One (Optional)
- > 0: Zero, One, or More (Optional Many)



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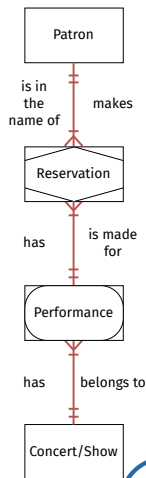
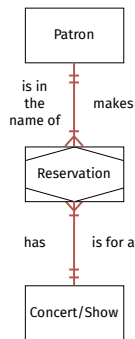
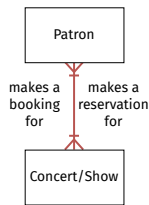
Types of Entities

- **Fundamental Entity (Standard Rectangle)**
 - Represents a basic object or concept (person, place, thing)
 - **Example:** STUDENT, DEPARTMENT, COURSE
- **Associative Entity (Rectangle with diamond inside corners)**
 - Links two or more other entities, often representing an event or transaction that depends on them
 - Used especially for M:M relationships
 - **Example:** An REGISTRATION links a COURSE and a STUDENT.
- **Attributive Entity (Rectangle with curved corners)**
 - Describes attributes, especially for repeating groups, of another entity
 - Dependent on a fundamental entity
 - **Example:** OFFERING might be an attributive entity holding specific semester and/or year for a COURSE



E-R Example: Concert Reservations

- Evolution of Understanding
 - Initial: Too simple?
 - Adding Associative: Better, Reservation links them
 - Adding Attributive: Handles multiple performances
- Shows how E-R models evolve as understanding deepens
- Attributes listed for each entity.
Underlined = Key/Searchable



Why Use E-R Diagrams Early?

- **Critical Analyst Task:** Start drawing E-R diagrams *early* in the project
- **Benefits**
 - Defines the **scope** and key entities of the problem
 - Helps understand the **business** itself (What are the key things they deal with?)
 - Helps ensure the **right problem** is being addressed
 - Provides a foundation for **data gathering** (interviews, observations, etc.)
 - Needs to be **confirmed** and **revised** as you learn more
- Don't wait for database design! It's an analysis tool first.

A Different Perspective: Use Case Modeling

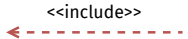
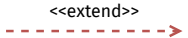
- **What:** A way to describe *what* a system does from the *user's* perspective, without detailing *how*
- **Focus:** System requirements and user interactions
- **Origin:** Unified Modeling Language (UML), but now widely used (SDLC, Agile)
- **Think:** Describing user goals and how the system helps achieve them

Use Case Basics: Actors and Use Cases

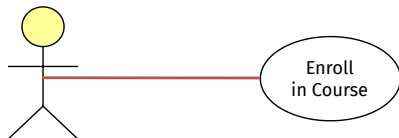
- **Actor:** Represents a role played by a user (human, another system, device) interacting with the system
 - Exists *outside* the system boundary
 - Can initiate or participate in a use case
 - Same person can be multiple actors (e.g., Employee, Customer)
 - **Symbol:** Typically a stick figure
 - **Primary Actors:** Initiate use cases, main users (e.g., Student)
 - **Supporting Actors:** Help maintain the system (e.g., Help Desk Staff)
- **Use Case:** Represents a specific goal or task an actor wants to achieve with the system
 - Describes a sequence of interactions
 - Should provide something of *value* to the actor
 - Named with Strong Verb-Noun (e.g., “Enroll in Course”, “Submit Order”)
 - **Symbol:** Typically an oval

How They Connect: Use Case Relationships

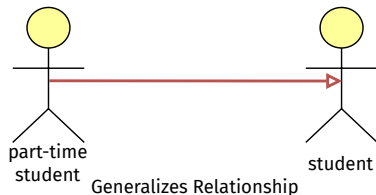
- **Behavioral Relationships:** Show how actors and use cases interact or relate to each other
- **Four Main Types**

Relationship	Symbol	Purpose
Communicates		Actor and Use Case interaction
Includes		One use case uses <i>another</i> common use case
Extends		One use case handles exceptions/variations for another
Generalizes		One actor/use case is more general version of another

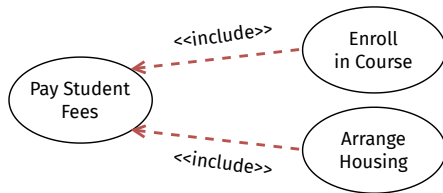
Use Case Relationships Explained



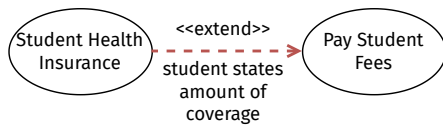
Communicates Relationship



Generalizes Relationship



Includes Relationship



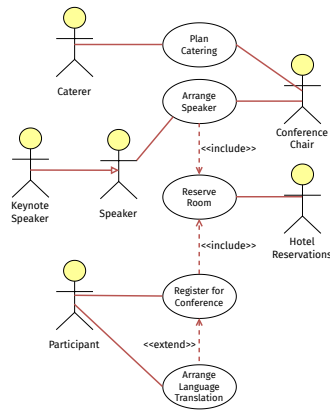
Extends Relationship

Developing Use Case Diagrams

- **Purpose:** Define system **scope** (actors are outside, use cases inside, communicates lines are boundaries)
- **Starting Point**
 - User interviews, JAD sessions, Agile stories → Ask “What should the system do for you?”
 - Existing Context-Level DFD
 - External Entities → Potential Actors
 - Data Flows → Potential Use Cases
- **Steps**
 - Identify **Actors** (review roles)
 - Identify high-level **Events** and **Primary Use Cases** initiated by actors
 - Determine **Variations/Alternative Paths** (these might become «extend» relationships or separate scenarios)
- **Goal:** Capture significant user interactions and goals. Keep it manageable (20-50 use cases for large systems)

Use Case Diagram Example: Conference Planning

- **Scenario:** System to help plan an academic conference
- **Actors**
 - Conference Chair
 - Participant
 - Speaker
 - Keynote Speaker
 - Hotel Reservations (Systems?)
 - Caterer



Beyond the Diagram: Use Case Scenarios

- **Need More Detail:** Diagrams show *what* happens, scenarios describe *how* (step-by-step)
- **Use Case Scenario:** A textual description detailing the sequence of steps within a specific use case
 - **Primary Path (Standard Flow/“Happy Path”):** Normal successful completion
 - **Alternative Paths:** Variations, error conditions, exceptions
- **Purpose:** Elaborate on the use case oval, providing necessary detail for design and development
- **Format:** Often uses a standardized organizational template

Use Case Levels (Cockburn's Altitude Metaphor)

- **Different Levels of Detail:** Not all use cases need the same granularity
- **Altitude Metaphor [2]**
 - **White (like clouds):** Highest level, enterprise-wide (4-5 total). E.g., "Manage Supply Chain"
 - **Kite:** High-level summary, business unit/dept. E.g., "Manage Reservations"
 - **Blue (Sea Level):** User goal level, most common. E.g., "Register Continuing Student" (takes 2-20 mins)
 - **Indigo (Fish):** Functional/Subfunctional detail. E.g., "Choose a Class", "Pay Fees"
 - **Black (Clam):** Most detailed, subfunction/implementation. E.g., "Validate Secure Logon"
- **Choose Level Appropriately:** Focus often on Blue level for user interaction understanding

Use Case Scenario Structure (Template)

- Common Sections

- Header

- Use Case Name and ID
 - System/Application Area
 - Actor(s)
 - Stakeholders (Interested parties, non-actors)
 - Level (Blue, Kite, etc.)
 - Description (Brief goal)
 - Trigger (Event starting the use case - external or temporal)

- Steps Performed

- Numbered sequence for the primary path
 - Information required/used for each step
 - Alternative paths/extensions

- Footer

- Preconditions (What must be true *before* starting)
 - Assumptions (About environment, user, tech)
 - Minimal Guarantee (Worst acceptable outcome)
 - Success Guarantee
 - (Desired outcome)
 - Outstanding Issues/Questions
 - **Optional:** Priority, Risk, Requirements Met

Use Case Scenario - Conference Registration

Use Case Scenario Example Details

- **Example** (Airline Reservation)
 - Shows detailed steps for the main path
 - Includes conditional logic within steps (e.g., “IF multiple airports...”)
 - Can reference alternative scenarios/extensions (Flight selection, Seat selection)
 - May include looping/iterative steps
- **Key:** Captures the flow and necessary logic clearly for developers

Use Case Scenario - Airline Reservation

Why Use Cases are Helpful

- **Communication:** Simple diagrams and stories are easy for non-technical users and stakeholders to understand
- **Requirements Focus:** Keeps focus on *what* the system needs to do for the user
- **Boundary Definition:** Clearly shows system scope and interactions
- **Traceability:** Links user needs to design elements and test cases
- **Versatility:** Useful across different development methodologies (SDLC, Agile, O-O)
- **Foundation:** Scenarios provide detailed steps for process modeling and test planning

Changing Perspectives: Levels of Management

- **Organizations aren't flat:** Management exists at different levels
- Why does this matter for Systems Analysis? → Different levels have different
 - Responsibilities and Goals
 - Decision-making styles
 - Information Needs
- Understanding these levels helps us design systems that serve *everyone* effectively
- **Think:** Different altitudes give different views of the landscape

The Three Tiers of Management

- **Typical Structure:** A three-tiered hierarchy
 - **Top:** Looking Outward and Forward
 - **Middle:** Bridging Strategy and Operations
 - **Bottom:** Managing Daily Activities
- **Think:** The Captain, Officers, and Crew of the ship



Level 1: Operational Control

- **Focus:** Direct supervision of daily operations and transactions
- **Decisions**
 - Based on predetermined rules
 - Predictable outcomes
 - Short-term horizon
 - **Examples:** Work scheduling, inventory control, shipping/receiving process control
- **Problem Type:** Structured, well-defined problems. Easy to identify
- **Alternatives:** Usually clear and easy to list
- **Decision Frequency:** Repetitive

Level 2: Middle Management

- **Focus**
 - Short-term planning and control
 - Allocating resources to meet objectives
- **Decisions**
 - How to best use resources (people, budget, equipment)
 - Monitoring performance against plans/standards
 - More variety than operational level
 - Examples: Developing departmental budgets, deciding on staffing levels, scheduling production for the next quarter
- **Problem Type:** Mix of structured and semi-structured problems
- **Alternatives:** May require more analysis than operational level
- **Decision Frequency:** Less repetitive than operational

Level 3: Strategic Management

- **Focus:** Long-term direction, looking outward at environment, setting overall goals
- **Decisions**
 - Define organizational mission and goals
 - Shape the organization's future (years ahead)
 - High uncertainty, broad scope
 - **Examples:** Entering new markets, developing new product lines, acquiring other companies, setting corporate policy
- **Problem Type:** Unstructured or semi-structured, often ambiguous. Difficult to identify.
- **Alternatives:** Often unclear, difficult to articulate or evaluate.
- **Decision Frequency:** Often unique, one-time decisions

Information Needs: Operational Control

- **Primary Need:** Internal, current performance data
 - Highly dependent on **real-time** or near real-time information
 - Repetitive, low-level detail
- **Moderate Need**
 - Past performance (for immediate comparison)
 - Period reports (daily/weekly summaries)
- **Example Systems:** Transaction Processing Systems (TPS), real-time dashboards showing production status, inventory levels

Information Needs: Middle Management

- **Primary Need:** Mix of internal information - current and historical
 - High need for **real-time** info (troubleshooting, control)
 - High need for **current performance** vs. **standards**
 - High need for **historical information** (trend analysis)
 - Need for **predictive information** and “what-if” scenarios (short-term planning)
- **Moderate Need:** Some external information (departmental competitors, industry trends)
- **Example Systems:** Management Information Systems (MIS), Decision Support Systems (DSS) providing summaries, exception reports, budget tracking, forecasting tools

Information Needs: Strategic Management

- **Primary Need:** External and Internal; Future-oriented
 - High dependence on **external information** (market trends, competitors, economy, regulations)
 - High need for **predictive information** and “**what-if**” scenarios (long-term strategy)
 - High need for **summarized, periodic information** (tracking overall progress)
 - Need for **risk assessment** (including security)
- **Moderate Need:** Summarized internal performance (the big picture)
- **Low Need:** Detailed, real-time internal operational data
- **Example Systems:** Executive Information Systems (EIS), DSS, Business Intelligence (BI) tools providing high-level dashboards, market analysis, long-range forecasting, risk modeling

Implications for IS Development

- **No Single System Fits All:** Information systems must be tailored to the target management level
- **Operational Level Needs:** Systems emphasizing transaction processing, accuracy, real-time monitoring, detailed reporting
- **Middle Management Needs:** Systems providing summaries, exception reporting, trend analysis, short/medium-term forecasting, DSS capabilities
- **Strategic Level Needs:** Systems integrating external data, supporting long-range planning, complex scenario modeling, high-level summarization (EIS/BI)
- **Analyst's Job:** Understand the user's level and specific decision-making context to determine appropriate system features and information presentation

Collaborative Design Across Levels

- **Concept:** Involving stakeholders from different levels (operational, middle, strategic) and departments in the system design process [3], [4]
- **Goal:** Create systems that meet diverse needs and gain buy-in
- **Challenges**
 - Power dynamics based on hierarchy
 - Differing perspectives and priorities
 - Ensuring relevant information flows to all participants
- **Opportunity:** Leveraging expertise from all levels (e.g., operational users know workflows, middle managers know constraints, strategic users know goals)
- **Key**
 - Structure collaboration carefully
 - Value diverse expertise
 - Manage information flow

Beyond Structure: Organizational Culture

- **What:** The shared values, beliefs, attitudes, and norms that shape behavior within an organization
 - The organization's "personality"
 - Often unwritten, but powerful
- **Why Care:** Culture significantly influences
 - How people work together
 - How decisions are *really* made
 - How readily new ideas (and systems!) are accepted
- **Think:** The invisible forces guiding how things get done

Not Just One Culture: Subcultures

- **Reality:** Organizations rarely have a single, uniform culture
- **Subcultures:** Smaller groups within the organization with their own distinct norms, values, and ways of communicating
 - Can form around departments, teams, roles, projects, even virtual teams
 - May align with, or sometimes conflict with, the “official” culture
 - Can compete for influence (“Our way is better!”)
- **Challenge:** Identifying and understanding these often overlapping and dynamic subcultures

Reading the Signs: Cultural Symbolism

- How to Perceive Culture/Subcultures? Look for symbols
 - Verbal Symbolism
 - Shared language, jargon, acronyms
 - Myths and stories about company history/heroes
 - Metaphors used (“We’re a family,” “We’re a war machine”)
 - Humor and inside jokes
 - Nonverbal Symbolism
 - Artifacts (awards, posters, logos)
 - Rites and ceremonies (parties, award events, retirement rituals)
 - Dress code (formal vs. casual)
 - Office layout and decoration (open plan vs. private offices, fancy vs. functional)

Culture's Influence on Information Systems

- **Major Impact:** Subcultures strongly influence IS requirements, adoption, and use
- **Potential Issues**
 - **Resistance to Change:** If a new system clashes with a subculture's values, workflow, or status
 - **Information Hoarding:** Subcultures might restrict information flow perceived as empowering rivals
 - **System Acceptance:** May be embraced by one group but rejected by another
 - **Determining Requirements:** Different subcultures may have conflicting needs or priorities for a system
- **Analyst Goal:** Understand subcultures to anticipate challenges and tailor implementation (e.g., targeted training)

Technology Shaping Culture: The Slack Example

- **Technology isn't neutral:** Tools can influence and shape organizational culture
- **Example:** Slack (Workplace Messaging App)
 - **Less Formal:** Than email, encourages quicker, more casual communication [5]
 - **Channel Structure:** Public (archived, searchable by team), Private (invitation-only), Direct Messages
 - **Potential Impacts**
 - Fosters collaboration and quicker communication
 - Can flatten hierarchy (easier access across levels?)
 - May create its own etiquette and norms (emoji use, response times)
 - Can potentially blur work/life boundaries
 - Shapes *how* colleagues interact and build relationships
- **Other Tools:** Similar impacts from collaborative platforms (Teams, Discord, etc.), project management software, video conferencing

Leveraging Culture Understanding

- **Be Observant:** Pay attention to verbal and nonverbal cues
- **Listen Actively:** Understand the stories, concerns, and perspectives of different groups
- **Identify Influencers:** Who are the key people respected within different subcultures?
- **Anticipate Resistance:** Where might clashes occur between the system and existing culture?
- **Tailor Approach**
 - **Communication:** Frame benefits relevant to specific subcultures
 - **Training:** Address specific concerns and workflows
 - **Implementation:** Consider pilot groups or phased rollouts based on cultural readiness
- **Goal:** Work *with* the culture(s) where possible, mitigate conflicts, and facilitate smoother system adoption

References I

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References II

- [5] Akumina, “2022 State of the Digital Workplace & Modern Intranet Report,” Akumina, Nashua, NH, Tech. Rep., 2022, White paper. [Online]. Available: <https://www.akumina.com/resources/2022-state-of-digital-workplace-report> (cit. on p. 47).